

Expressions Workshop: Distribute, Combine, Factor

Answer Key

Grade 6–7 Expressions and Equations

Quick Answers

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|---------------|--------------|---------------|---------------|
| 1. $4x + 28$ | 4. $5m + 7$ | 7. $5x + 18$ | 10. $6x - 15$ |
| 2. $15x - 10$ | 5. $5x + 12$ | 8. $12n - 18$ | |
| 3. $9x + 11$ | 6. $6x + 15$ | 9. 78 | |
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1. Area model for $4(x + 7)$. The height 4 multiplies *each* piece of the base, because the two little rectangles together make the whole rectangle:

$$4 \cdot x = 4x \quad \text{and} \quad 4 \cdot 7 = 28$$

Total area = $4x + 28$ square metres.

$4x + 28$

The model is the reason the rule works: splitting the base splits the area.

2. Five rooms of $(3x - 2)$ minutes. Five equal boxes means 5 copies added:

$$5(3x - 2) = 5 \cdot 3x - 5 \cdot 2 = 15x - 10$$

The minus sign belongs to the 2, so the second product is subtracted. Total time =

$15x - 10$

 minutes.

3. Combine $7x + 3 + 2x + 8$. Sort first: the x -terms are $7x$ and $2x$; the number terms are 3 and 8.

$$(7x + 2x) + (3 + 8) = 9x + 11$$

Only like terms combine — $9x$ and 11 count different things (rows of tiles versus single tiles), so the answer keeps both.

$9x + 11$

4. Combine $8m + 5 - 3m + 2$. Each term keeps the sign in front of it, so the m -terms are $+8m$ and $-3m$:

$$(8m - 3m) + (5 + 2) = 5m + 7$$

Tia has

$5m + 7$

 markers. Check with $m = 10$: the original gives $80 + 5 - 30 + 2 = 57$, and $5(10) + 7 = 57$ ✓.

5. Simplify $3(x + 4) + 2x$. Distribute over the three curved sections first, then combine:

$$\begin{aligned} 3(x + 4) + 2x &= 3x + 12 + 2x \\ &= (3x + 2x) + 12 \\ &= 5x + 12 \end{aligned}$$

Total path length =

$5x + 12$

 metres. Note $2x$ is not inside the parentheses, so it is never multiplied by 3.

6. Factor $6x + 15$. List the factors: $6x = 2 \cdot 3 \cdot x$ and $15 = 3 \cdot 5$. The greatest common factor is 3, so the height of the model is 3 and the two base pieces are $6x \div 3 = 2x$ and $15 \div 3 = 5$:

$$6x + 15 = 3(2x + 5)$$

Area as a product:

$$3(2x + 5)$$

. Distributing back gives $6x + 15$ ✓ — factoring is the area model run backwards.

7. Simplify $4(2x + 3) - 3(x - 2)$. Distribute both products, and give the second one the subtraction sign it owns:

$$4(2x + 3) = 8x + 12$$

$$-3(x - 2) = -3x + 6$$

$$8x + 12 - 3x + 6 = (8x - 3x) + (12 + 6) = 5x + 18$$

Badges kept = $5x + 18$. The trap is $-3 \cdot (-2)$: a negative times a negative is $+6$, so the constants grow rather than shrink.

8. Factor $12n - 18$. $12n = 2 \cdot 2 \cdot 3 \cdot n$ and $18 = 2 \cdot 3 \cdot 3$, so the greatest common factor is 6. Dividing each piece by 6 gives $2n$ and 3:

$$12n - 18 = 6(2n - 3)$$

Remaining area as a product:

$$6(2n - 3)$$

. Pulling out only 2 or only 3 is not wrong arithmetic, but it is not the *greatest* common factor, so the model would still have a shared factor left inside.

9. Total tutoring cost $5(x + 3) + 4x$ **dollars.** (a) Distribute, then combine like terms:

$$5(x + 3) + 4x = 5x + 15 + 4x = 9x + 15$$

(b) Substitute $x = 7$ into the simplified expression:

$$9(7) + 15 = 63 + 15 = 78$$

The cost is

$$78$$

dollars. Substituting into the original form is a good check: $5(7 + 3) + 4 \cdot 7 = 50 + 28 = 78$ ✓ — equivalent expressions must agree at every value of x .

10. Challenge — Jamal's error. Jamal multiplied the $2x$ by 3 but left the -5 untouched. The distributive property says the factor outside multiplies *every* term inside, because $3(2x - 5)$ means three copies of the whole quantity $2x - 5$:

$$3(2x - 5) = 3 \cdot 2x - 3 \cdot 5 = 6x - 15$$

Correct expanded expression:

$$6x - 15$$

. Test the two at $x = 1$: the correct form gives $6 - 15 = -9$, which matches $3(2 - 5) = -9$; Jamal's $6x - 5$ gives 1, so his version is a different expression. *Note for the grader: the written explanation is judged by you — only the corrected expression is machine-verified.*